

## Education

- Oct. 2022 - **PhD in computational and theoretical neuroscience at ENS-PSL**, *Learning dynamics in biologically plausible neural networks: theory and data-driven modeling*. Under the supervision of Alex Cayco Gajic and Jonathan Kadmon. Recipient of a selective scholarship (CDSN).
- Developed a biologically inspired learning algorithm and evaluated its efficiency relative to classic gradient descent through large-scale simulations on a local cluster.
  - Conducted statistical analysis of high-dimensional electrophysiological brain recordings to investigate the role of cerebellar signals carrying sensorimotor information in motor preparatory activity during adaptation.
- 2021-2022 **Graduate student (M2 MVA : Mathematics, Vision and Learning)**, Université Paris-Saclay.
- *Classes*: Algorithms for speech and natural language processing; Computational statistics; Convex optimization and applications in machine learning; Information and Complexity; Modeling in neuroscience; Theoretical neuroscience; Topological data analysis for imaging and machine learning; Responsible machine learning.
- 2020-2021 **Graduate student (M2 ICFP) : Theoretical physics track and École normale supérieure diploma (DENS)**.
- *Classes*: Machine Learning; Advanced topics in Markov-chain Monte Carlo; Advanced Statistical Physics and New Applications; Numerical Physics and Machine Learning; Complex systems : from physics to social sciences; Information, inference, networks : from statistical physics to big biological data; Biophysics; Quantum Field Theory; General Relativity.
- 2018-2019 **Graduate student (M1 ICFP and DENS)**, École normale supérieure de Paris.
- 2017-2018 **Bachelor in physics and DENS**, École normale supérieure de Paris, admitted through competitive examination (rank 25/1137).
- 2014 - 2017 **Preparatory classes for the Grandes Ecoles, Physics and chemistry track**, Lycée Aux Lazaristes, Lyon.

## Teaching experience

- 2024-2025 **Teaching assistant**, *Designed and delivered the tutorials for the course 'Mathematical tools for data science, machine learning, and statistical modeling' taught by Prof. Alex Cayco-Gajic.*

## Publications

- Sept. 2025 **Curl Descent: Non-Gradient Learning Dynamics with Sign-Diverse Plasticity**, Hugo Ninou, Jonathan Kadmon, N. Alex Cayco-Gajic, **Neurips 2025 Spotlight**.
- March 2025 **Cerebellar output shapes cortical preparatory activity during motor adaptation**, Sharon Israely\*, Hugo Ninou\*, Ori Rajchert, Lee Elmaleh, Ran Harel, Firas Mawase, Jonathan Kadmon, Yifat Prut, \*These authors contributed equally, **Nature Communications**.

## Awards

- 2025 **NeurIPS 2025 Scholar Award**.
- 2022 **Awarded a competitive PhD scholarship (CDSN) by the Ecole normale supérieure**.

## Conferences and presentations

- Mar. 2026 **ZenkeLab, Friedrich Miescher Institute for Biomedical Research**, *Invited talk 'Learning Beyond the Gradient: The Computational Advantage of Adversarial Plasticity'*.
- Feb. 2026 **Flatiron Institute**, *Invited talk 'Learning Beyond the Gradient: The Computational Advantage of Adversarial Plasticity'*.

- Dec. 2025 **NeurIPS 2025**, *Spotlight poster 'Curl Descent: Non-Gradient Learning Dynamics with Sign-Diverse Plasticity'*.
- Oct. 2025 **Bernstein 2025**, *Poster 'Curl Descent: Non-Gradient Learning Dynamics with Sign-Diverse Plasticity'*.
- July 2025 **OCNS 2025**, *Poster 'Curl Descent: Non-Gradient Learning Dynamics with Sign-Diverse Plasticity'*.
- Apr. 2025 **ShineLab, University of Sydney**, *Talk on 'Cerebellar output shapes cortical preparatory activity during motor adaptation'*.
- Jan. 2025 **Najafi Lab SYBECO journal club, Georgia Tech**, *Talk on 'Cerebellar output shapes cortical preparatory activity during motor adaptation'*.
- March 2024 **COSYNE 2024**, *Poster 'Cerebellar output shapes cortical preparatory activity during motor adaptation'*.

## Research experience

- May - Sept. 2022 **Hebrew University of Jerusalem**, *Analyzing neural dynamics in the cortico-cerebellar pathway of non-human primate brain during learning. Supervision by Jonathan Kadmon and Yifat Prut.*
- High-dimensional data analysis on electrophysiological neural recordings.
  - Implementation of dimensionality reduction techniques (Targeted dimensionality reduction, PCA) for analysis of neural trajectories through adaptation in the motor cortex.
- April 2021 - July 2021 **ENS-PSL**, *Inference of learning rules in biologically plausible recurrent neural networks. Supervised by Rémi Monasson.*
- Statistical data extraction and analysis on calcic recordings of zebra fish whole-brain activity.
  - Modeling this brain activity and its learning process based on a Recurrent neural network (RNN) and General linear models (GLM).
- Dec. 2019 - May 2020 **LaDHyX, École Polytechnique - Capital Fund Management**, *Studying and modeling emergence phenomena in price heterogeneities. Supervised by Michael Benzaquen and Alan Kirman.*
- Data extraction and statistical analysis (PCA).
  - Comparison between usual statistical physics model (Ornstein-Uhlenbeck) to fit the data and conception of an agent based model having phases associated to the different observed behaviors.
- Aug. - Nov. 2019 **Canada France Hawaii Telescope**, *Development of a convolutional neural network for detecting parasite sources in the data harvested by the SITELLE spectrometer. Supervision by Simon Prunet and Laurie Rousseau-Nepton.*
- Making of training sets and testing sets through defect generation on clean images.
  - Conception and training of a deep neural network for computer vision (Convolutional neural network, CNN) able to detect and label the different types of image defects.
- Febr.-July 2019 **University of California San Diego**, *Synthesis of self-propelled swimmers and studying of emergence phenomena in their collective behaviour. Supervision by Jérémie Palacci.*
- Participation in developing an experimental setup together with exploring unknown phenomena.
  - Developement of a synthesis protocol for self-propelled diffusio-phoretic swimmers able to resist high constraints.
- 2018-2019 **Lycée Michelet and Lycée Saint-Louis**, *Oral examiner (prépa PCSI and MP\*).*
- June 2018 **Institut Lumière Matière**, *Electro-cinetic flows in wetting films. Supervision by Anne-Laure Biance.*
- Making of an experimental setup enabling the measure of electro-cinetic flows in a wetting film.

## Other

- Feb. 2025 **Artificial Intelligence Action Summit, Paris**, *Took part in the PSL student assembly as a participant.*